WESTERN MICHIGAN UNIVERSITY

**College of Engineering and Applied Sciences**

**Course Name: Lab. Intro. To Microprocessors**

**Course Number:** ECE 2510

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**Lab #7 prelab**

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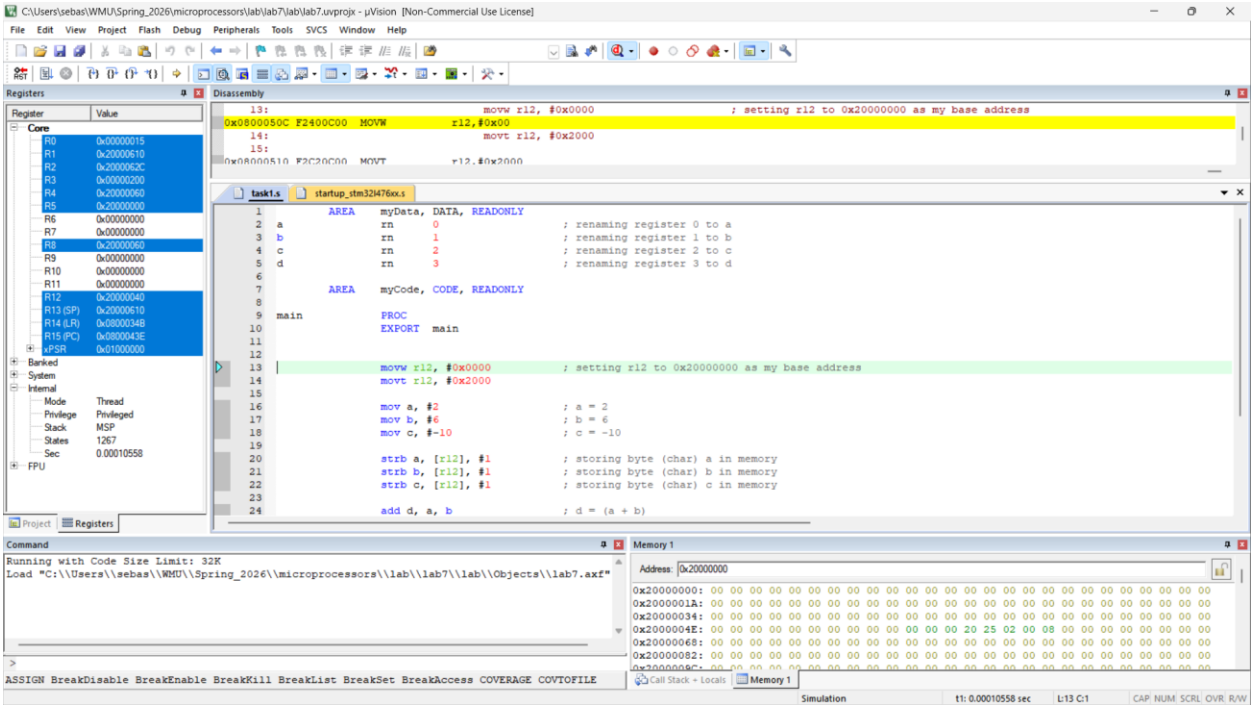
# Task 1

# Initial state of program:



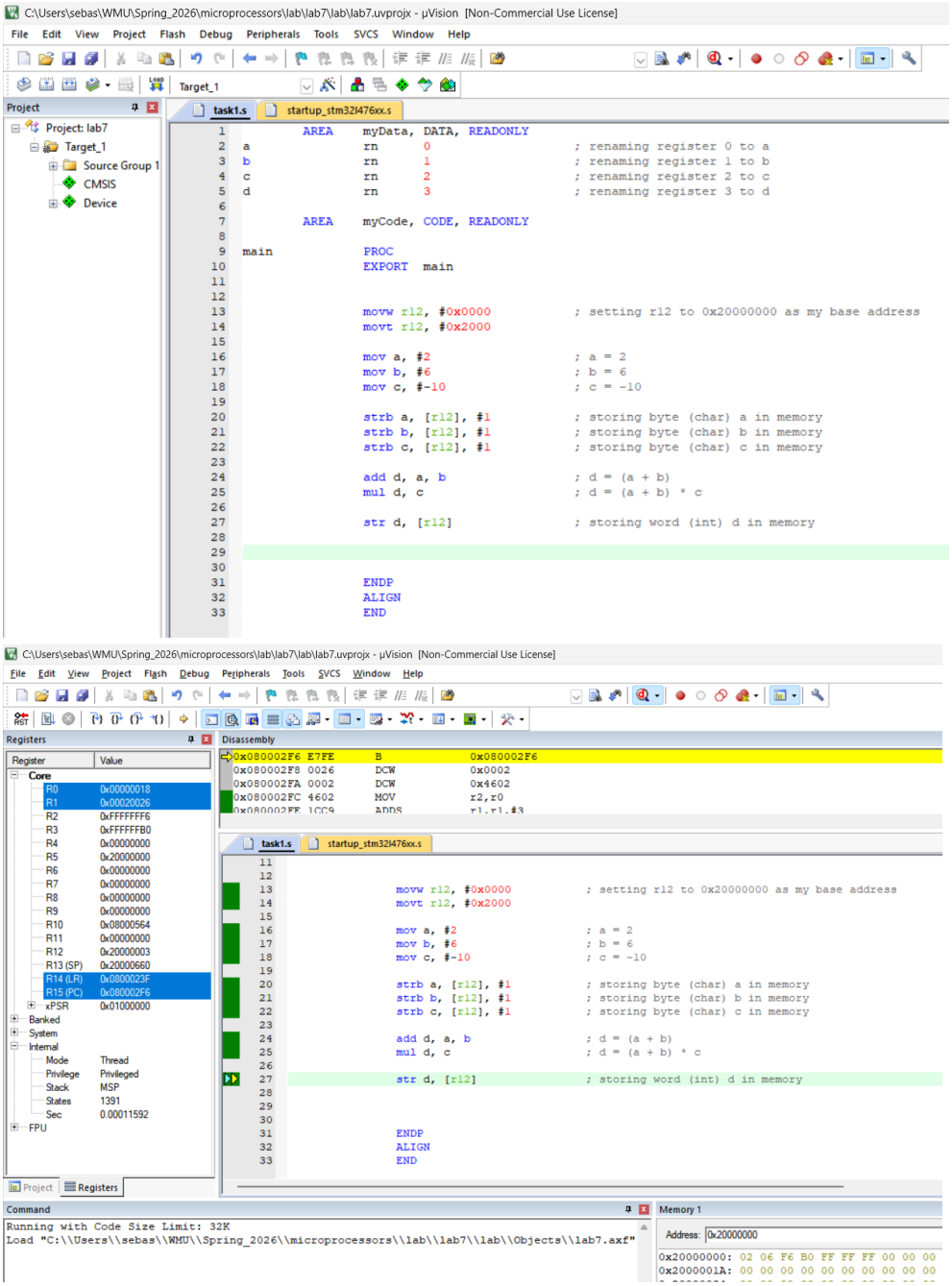


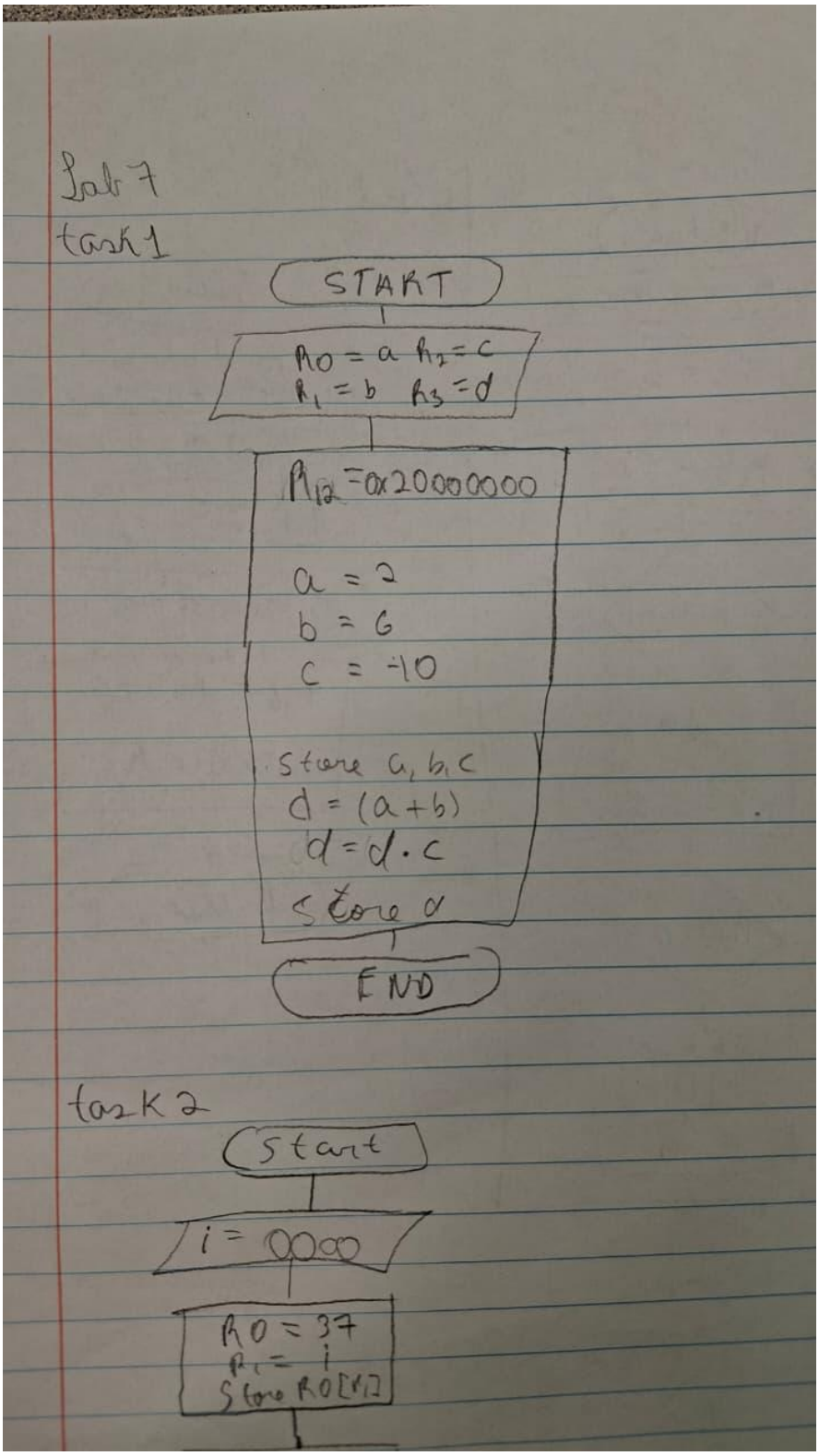

*Figure 1: Task 1 results*

E-Cycles = Time/Clock

Assyming Clock = 4MHz -> 250 nSec

E-Cycle s= 0.00011592 secs / 250 nSec = 463.68 = 464 cycles

Flowchart


Lab 7
task 1
START
R0 = a R2 = c
R1 = b R3 = d
R12 = 0x20000000
a = 2
b = 6
c = -10
Store a, b, c
d = (a+b)
d = d · c
Store d
END
task 2
Start
i = 0000
R0 = 37
R1 = i
Store R0[R1]

**Task 2**

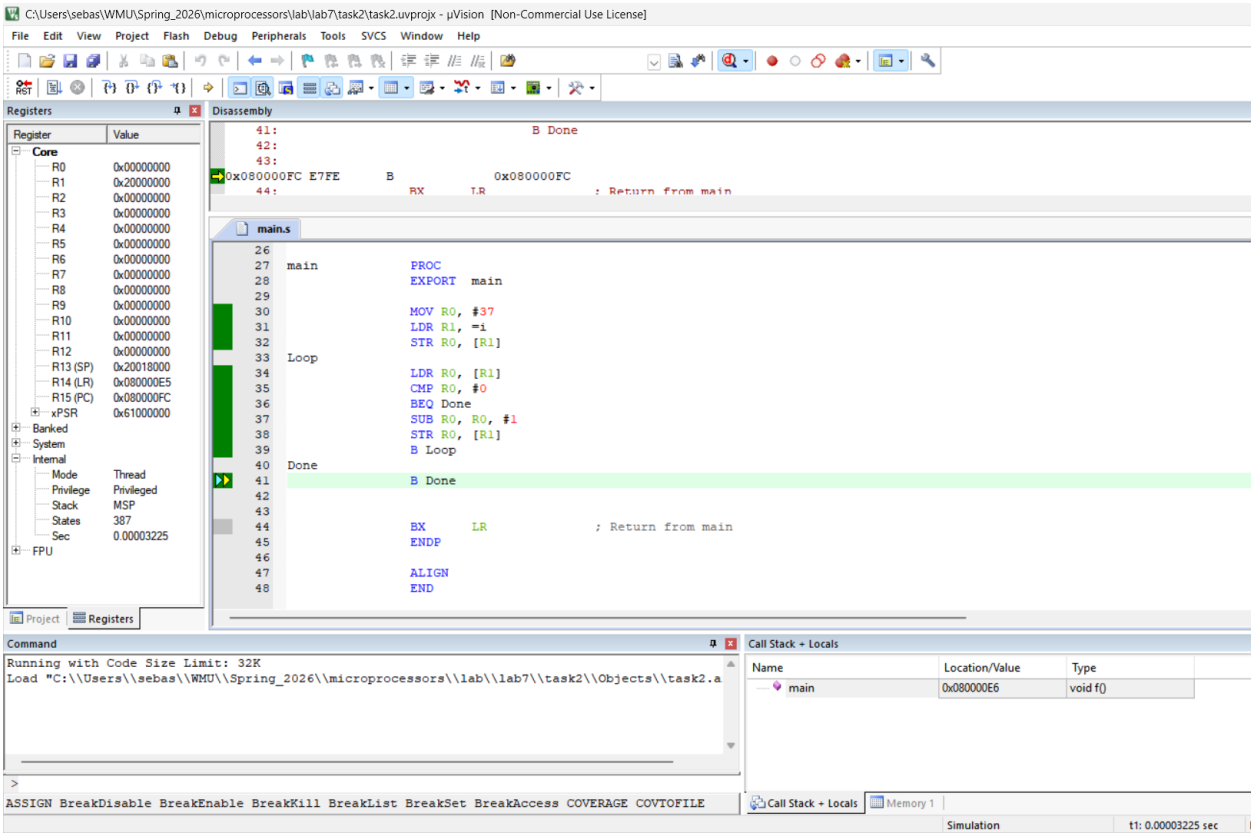
```
AREA myData, DATA, READWRITE
i SPACE 4

AREA myCode, CODE, READONLY
EXPORT __main
ENTRY

__main
        MOV R0, #37
        LDR R1, =i
        STR R0, [R1]
Loop
        LDR R0, [R1]
        CMP R0, #0
        BEQ Done
        SUB R0, R0, #1
        STR R0, [R1]
        B Loop
Done
        B Done

END
```

(prelab work) counting cycles starting at _main

(1 + 2 + 2 + [(loop label) 2 + 1 + (BEQ not taken) 1 + 1 + 2 + (3)] * 36 times + (CMP) 1 + (BEQ taken) 1 + 1 ) cycles = 368 cycles

Time in Seconds = 368 cycles * 250 nanoseconds = 92 miliseconds

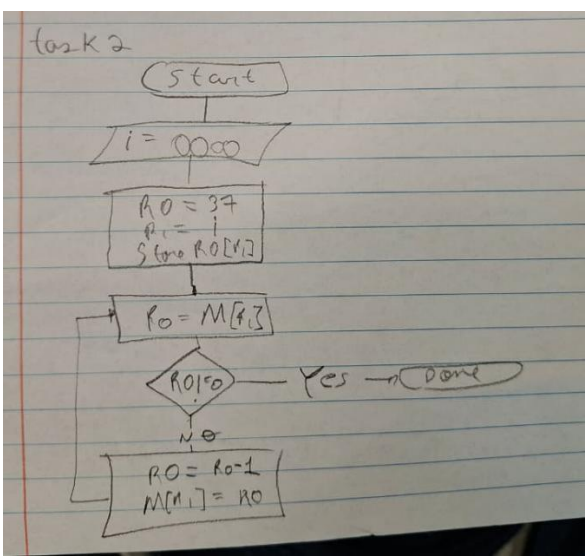
My results:




Sec: 0.00003225,

E-Cycles = 0.00003225 / 250 nSec = 129 cycles

Flowchart



**Task 3**

Determine the outer and inner loop counts in the ARM assembly code below to produce a 0.5-second delay (F = 4 MHz).

```
AREA myCode, CODE, READONLY
EXPORT __main
ENTRY

__main
        PUSH {R4, R5, LR}
        MOV R4, #?
loop1
        MOV R5, #?
loop2
        SUBS R5, R5, #1
        BNE loop2
        SUBS R4, R4, #1
        BNE loop1

        POP {R4, R5, PC}
END
```

Answer the following questions:

1- How many cycles you will need to create a delay of 0.5 seconds (Prelab)?

Target delay = 0.5 seconds = 0.5 seconds / 1*10^-9 = 500000000 nanosecs

Total cycles needed = 500000000 nanosecs / 250 nanosecs/cycle = 2x10^6 cycles

Total delay = Outer * Inner * 4 * 250 ns

500,000,000 ns  / 1000 ns = Outer * Innner

Outer * Inner = 500,000 ns

Let Outter = 1000

Inner = 500

2- How many iterations do you need to execute the loop on slide 5 to have a 0.5-second delay (Prelab)?

I need 500,000 Loop counts

3- Does the answer for 2 fit in one ARM register? If not, what do you need to do to overcome the problem? (Prelab)

no, I will do it in two loops, one nested inside the other, inner loop 500 iterations, outer loop 1000 iterations

4- Were you able to get the exact same cycles in delay? Why or why not?

yes,

5- What is the total delay time that you were able to create?

Half a second

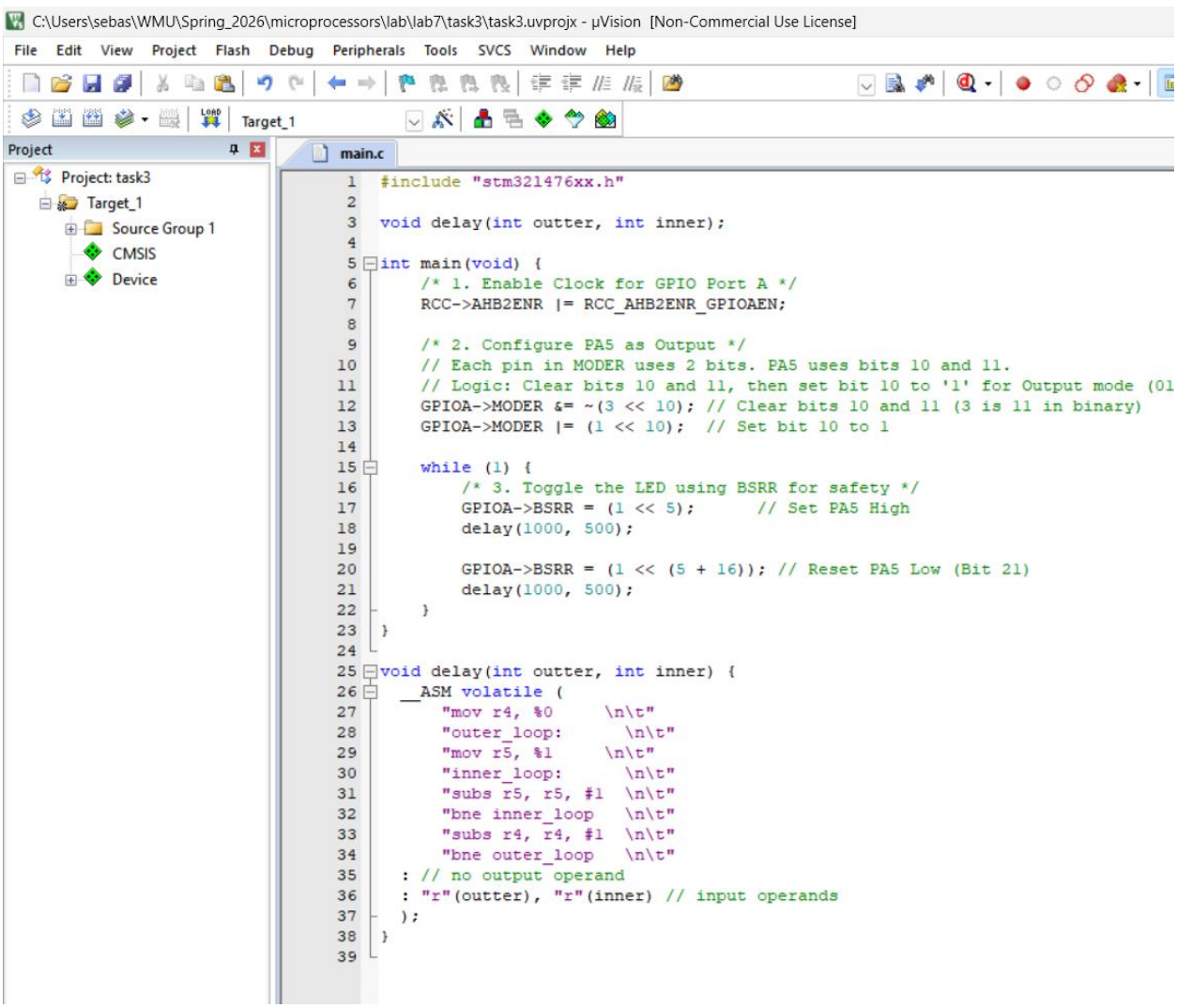



*Figure 2: My C code for task 3*

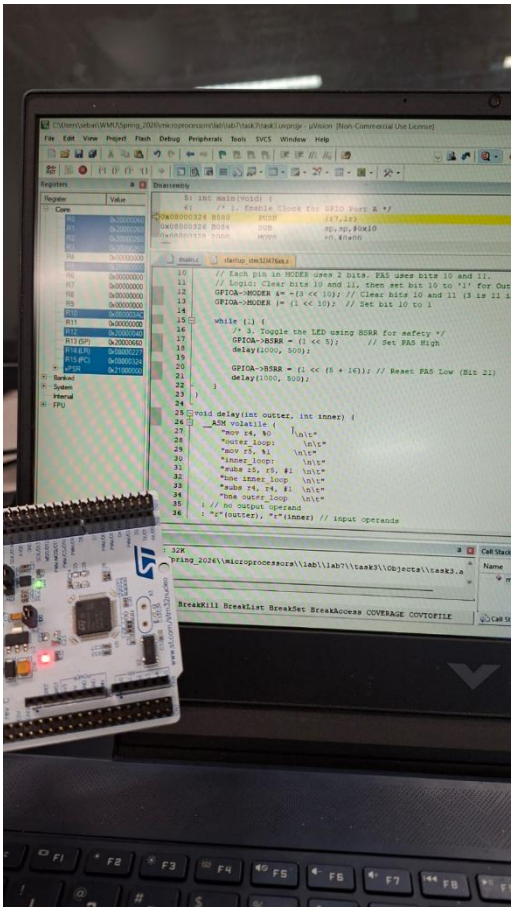


*Figure 4: Code running on board LD2 turned on*

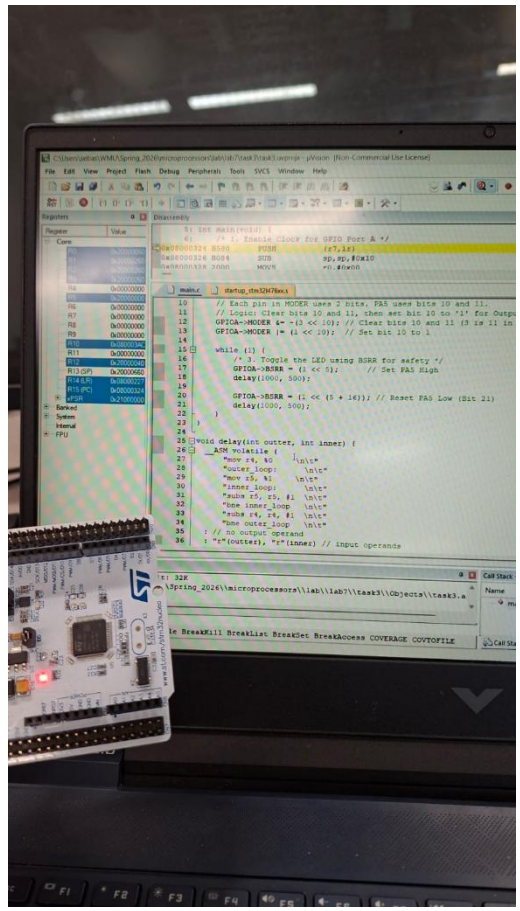
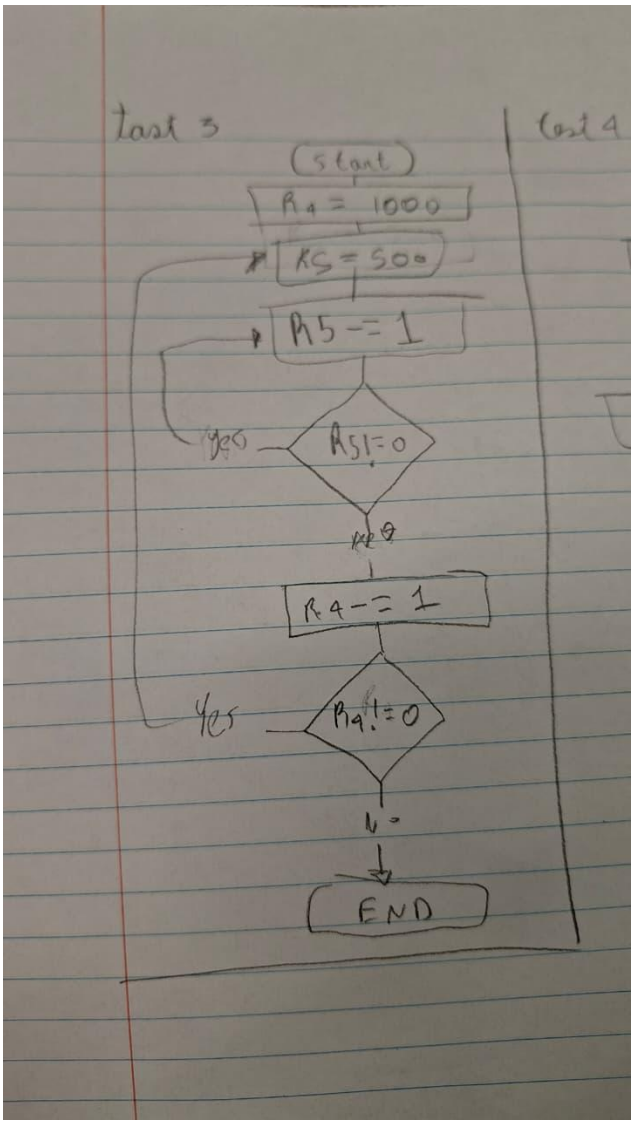
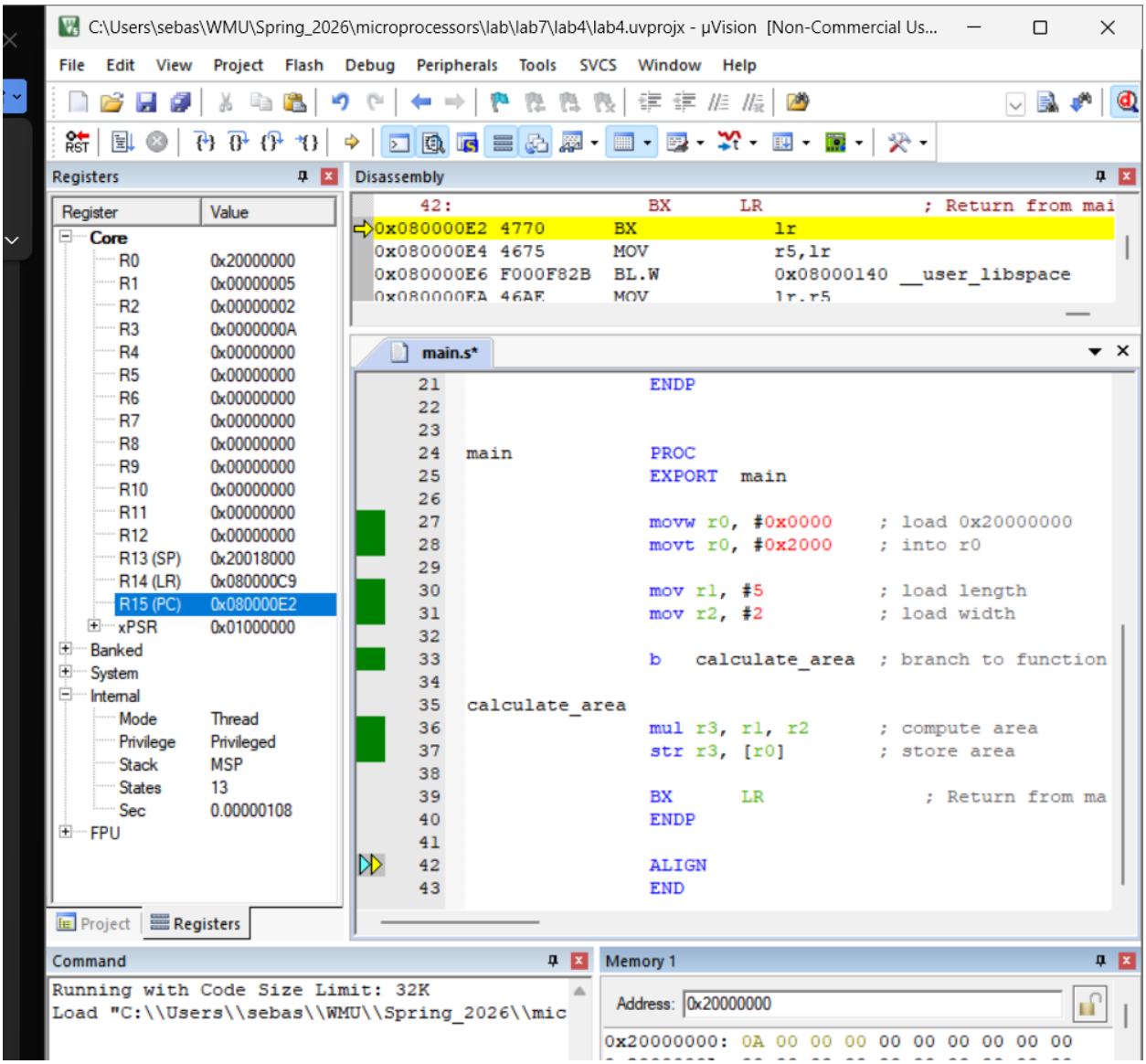
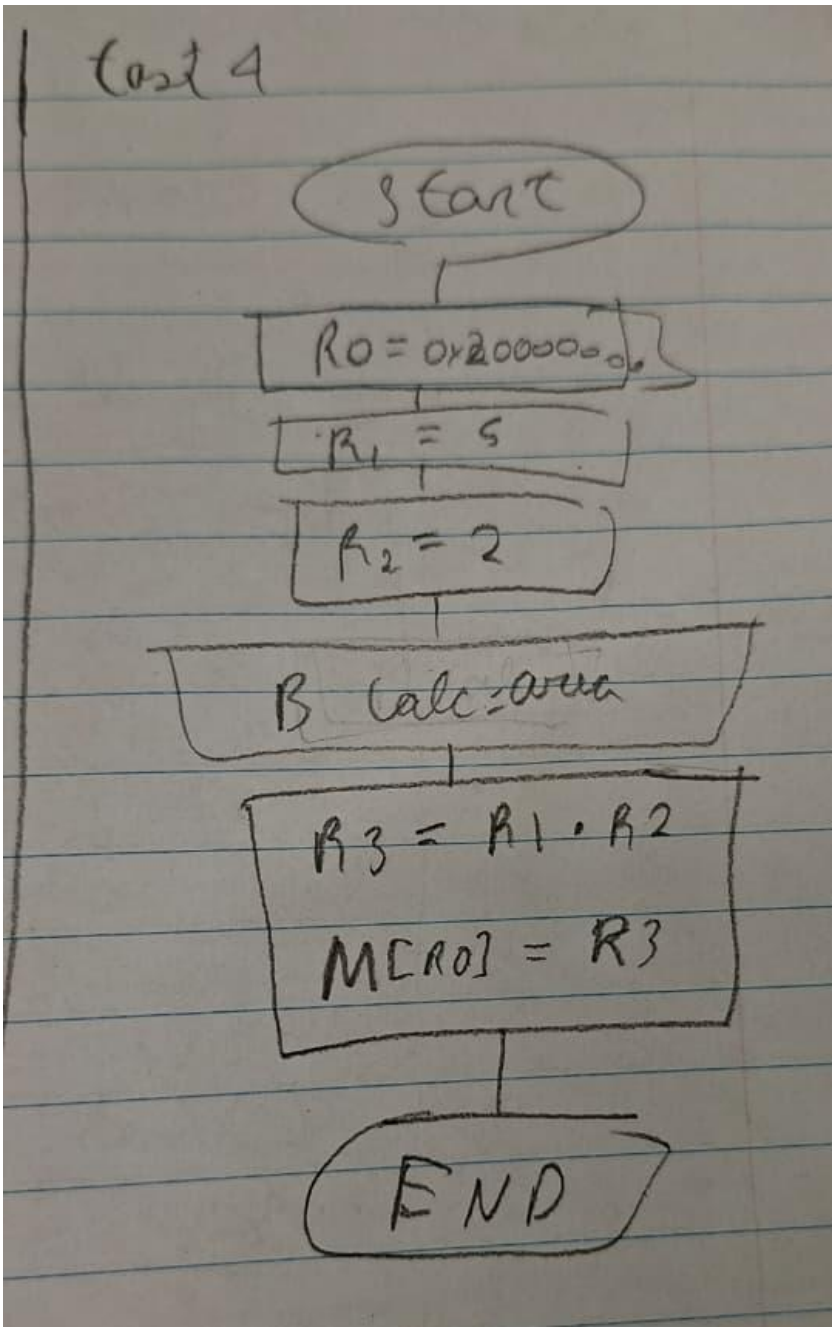


*Figure 3: Code running on board LD2 turned off*

Flowchart



Task 4


C:\Users\sebas\WMU\Spring_2026\microprocessors\lab\lab7\lab4\lab4.uvprojx - µVision [Non-Commercial Us...
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x20000000
R1 0x00000005
R2 0x00000002
R3 0x0000000A
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E2
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 13
Sec 0.00000108
FPU
Project Registers
Disassembly
42: BX LR ; Return from mai
0x080000E2 4770 BX lr
0x080000E4 4675 MOV r5,lr
0x080000E6 F000F82B BL.W 0x08000140 __user_libspace
main.s*
ENDP
main PROC
EXPORT main
movw r0, #0x0000 ; load 0x20000000
movt r0, #0x2000 ; into r0
mov r1, #5 ; load length
mov r2, #2 ; load width
b calculate_area ; branch to function
calculate_area
mul r3, r1, r2 ; compute area
str r3, [r0] ; store area
BX LR ; Return from ma
ENDP
ALIGN
END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\WMU\\Spring_2026\\mic
Memory 1
Address: 0x20000000
0x20000000: 0A 00 00 00 00 00 00 00 00 00

Flowchart